

TAYYAB HUSSAIN

AI ENGINEER

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SUMMARY

Aspiring AI Engineer with hands-on experience building production-grade multi-agent LLM systems — LangGraph orchestration, RAG pipelines, and full-stack AI products shipped under real constraints. Open to entry-level and internship opportunities in AI/ML engineering. Kaggle Notebooks Master and 2nd-place finisher in a 100+ team international hackathon.

TECHNICAL SKILLS

AI / LLM Engineering: LangChain, LangGraph, RAG Pipelines, Multi-Agent Systems, Prompt Engineering
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Keras, NumPy, Pandas, OpenCV, Hugging Face Transformers
Backend & APIs: FastAPI, REST APIs, Python (Advanced), SQL
Frontend: React, Tailwind CSS
Data & Vector Databases: ChromaDB, Pinecone, PostgreSQL, Redis
Deployment & Tools: Docker, Git, GitHub, Jupyter, Model Deployment, API Optimization

PROJECTS

- FashionOS — Multi-Tenant Fashion Retail SaaS** | *LangGraph, FastAPI, FastMCP, React/Vite, PostgreSQL, Clerk*
- Multi-tenant retail platform with custom FastMCP servers for Shopify (GraphQL Admin API), Meta (WhatsApp Cloud + Ads API), and Apify (Instagram/TikTok scraping) — dynamic per-brand credential isolation across all integrations.
 - LangGraph "Batch + Serve" agentic pipeline: daily scheduled graph (MCP data collection → LLM analysis → PostgreSQL) paired with an on-demand ReAct agent; integrates Google Trends MCP for real-time trend intelligence.
- MediTwin AI — Clinical Decision Support System** | *LangGraph, FastAPI, Gemini 2.5 Flash, Docker, ChromaDB, XGBoost, React*
- 10-agent clinical AI system (LangGraph StateGraph) across 10 Dockerized microservices — differential diagnosis (RAG + ChromaDB), drug safety (FDA MCP), chest X-ray triage (EfficientNetB0, 95.5% acc), and digital twin simulation (XGBoost).
 - ReAct agent with 8 tools, PostgreSQL memory, SSE streaming; outputs FHIR R4 bundles and SOAP notes with A2A compliance in under 12 seconds.
- AINutritionChef** | *LangGraph, Gemini 2.5 Flash, FastAPI, PostgreSQL, Redis, React*
- Multi-agent pipeline (health goal → recipe generation → macro validation → allergen substitution) with RAG retrieval, Redis caching, food image recognition, 7-day meal planning, and JWT auth.
- Agri Advisory System** | *LangGraph, Gemini 2.5 Flash, FastAPI, Celery, Redis, PostgreSQL, WhatsApp Cloud API*
- WhatsApp crop photo → Gemini Vision disease diagnosis → bilingual Urdu/English advice in under 30 seconds; Celery + Redis async pipeline, nightly NDVI/weather refresh, and a React field-monitoring dashboard.
- Debate AI System** | *LangGraph, Claude API, FastAPI, PostgreSQL, React*
- 6-agent LangGraph debate trainer (Moderator, Argument Analyzer, Researcher, Socratic Questioner, Devil's Advocate, Growth Tracker) with SSE streaming, fallacy detection, argument scoring, and PostgreSQL session history.

EDUCATION

Bahauddin Zakariya University

Bachelor of Science in Information Technology (CGPA: 3.40)

- Relevant Coursework: Data Structures & Algorithms, Linear Algebra

2022 – present

Multan, Pakistan

ACTIVITIES & RECOGNITION

Kaggle Notebooks Master	2024
• Authored technical notebooks on neural network architectures, gradient boosting, RAG systems, and EDA — accumulating thousands of views and upvotes from the global ML community.	
2nd Place — International AI Hackathon (100+ teams)	2025
• Delivered a complete end-to-end AI product within a 72-hour sprint, recognized for technical depth and product design.	
Active AI Hackathon Competitor	2025 – Present
• Regularly compete in international AI hackathons, building production-grade multi-agent and RAG systems under sprints ranging from a few days to several weeks against global teams.	